$\boldsymbol{\delta}$: The Formal Unification of Mathematics and Physics from Distinction

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Abstract

We introduce δ, pronounced *distinction*, as a formal foundation in which mathematics and physics factor through a single primitive act. Mathematics is the permission theory of distinction: what distinction allows structurally. Physics is the cost theory of distinction: what distinction costs when comparison becomes measurable. Both readings operate on the same carrier. The historical split between formal structure and physical law is therefore a split in presentation, not in source.

Starting from a single primitive (the act of marking a difference), we construct the finite δ kernel and derive the first complete number tower. The construction proceeds: distinction acts compose into traces; trace-level equivalence yields quotient structure; repetition length yields base-neutral natural numbers; balanced pairs yield integers; cross-multiplied ratios yield rationals. We prove the arithmetic laws (associativity, commutativity, distributivity) for each level, construct internal order, an exact signed-orbit comparator, absolute value, order transport under addition and negation, divisibility, primality, Euclidean quotient-remainder, gcd, coprimality, and a rational field. A Cauchy completion surface and a rational J-cost bridge connect the construction to the existing uniqueness theorem for the canonical reciprocal cost $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Every construction carries a strength tag (δ-only, δ + trace closure, δ + choice, or classical extension) so that the proof strength of any downstream claim is immediately auditable. All results through the rational field are δ-only: no axiom of choice, no law of excluded middle, no axiom of infinity. The present formal corpus proves a conditional universal-foundation certificate and isolates, as exact theorems, what is forced versus what is a gauge at the one load-bearing joint: the forcing of the cost function J. The native arbitrary-cost uniqueness blocker is resolved as a negative. On the discrete rational carrier J is *not* forced: orientation on each prime axis is an independent free choice (we construct ratio characters that fix every other prime and flip one). On the continuous completion the algebraic laws (reciprocal symmetry, the composition law, normalization, continuity) force only the one-parameter *family* $F_\lambda(x) = \frac{1}{2}(x^\lambda + x^{-\lambda}) - 1$, which is exactly the orbit of J under the automorphism group of the positive reals under multiplication. The calibration $\lambda^2 = 1$ selects J within that family; the value of that calibration is a multiplicative-automorphism gauge that δ does not fix. The honest conclusion is therefore: δ forces the cost *form*, and a unit calibration fixes the gauge that selects J. We package this stratification as a single machine-checked object.

Three companion papers supply the continuous carrier and geometric realization. The functional-equation paper proves the canonical reciprocal cost on positive ratios [8]. The arithmetic paper recovers number from the Law of Logic on the continuous carrier [9]. The recognition-geometry paper connects event quotients to comparison [10]. The present paper sits below those results by constructing the carrier surface from distinction itself. The current formal state proves a conditional universal-foundation certificate: kernel, logic, arithmetic, rational field, real-completion surface, formal-system embedding, admissible-foundation inevitability, recognizer cost bridge, and the resolved cost-joint stratification all compose. The cost-forcing question is settled (as a negative): δ forces the cost form and a unit calibration fixes the gauge that selects J. The only remaining open item before an unqualified theorem is parsing historical external foundations into the formal-system interface.

Contents

1 Introduction

1.1 The inherited split

This paper constructs a formal system in which mathematics and physics share a single primitive. The primitive is an act of distinction, written δ. From it we derive the natural numbers, integers, rationals, order, divisibility, primality, Euclidean structure, a rational field, and a cost function whose *form* is forced by symmetry and composition constraints, with the unit calibration a gauge choice made precise in Section 7 rather than an additional consequence of δ. The cost function is the one that Recognition Science uses to derive physical constants, including the fine-structure constant α, from first principles. Because the mathematical carrier and the physical cost function both originate from δ, the split between mathematics and physics reduces to a split in the question asked, not a split in the subject.

The division between the two fields is old enough that it feels natural. It is not. It is an accident of intellectual history, hardened by centuries of separate departments, separate journals, separate notations, and separate hiring committees. Mathematics begins with sets, types, universes, judgments, or categorical objects. Physics begins with states, fields, particles, manifolds, actions, Hamiltonians, and empirical constants. Both traditions built enormous structures from their chosen primitives, and both traditions succeeded.

But the success concealed a shared premise. Every mathematical foundation presupposes the ability to distinguish. A set is defined by what belongs to it and what does not. A type is defined by which terms inhabit it and which do not. A proposition has a truth value only because true can be distinguished from false. On the physics side, the same act appears under different names. A measurement distinguishes one outcome from another. A field configuration has content because it differs from other configurations. A state has identity only because alternatives exist. The act of distinction is there in every axiom system, every measurement, every proof. It is treated as background, too obvious to formalize.

δ takes that background and makes it the foreground. One primitive: the act of distinction. A finite repetition of the act is a *trace*. Trace equivalence gives sameness. Trace inequivalence gives difference. Repetition length gives number. Balance gives sign. Cross-comparison gives ratio. Measured comparison gives cost. Everything in the list is derived, not assumed.

Main claim. Mathematics and physics are formally one because both factor through distinction. Mathematics is the permission reading of δ. Physics is the cost reading of δ. Recognition Science is the program that proves, tests, and builds the consequences of that fact.

1.2 Contributions

This paper makes five concrete contributions.

1. It defines δ as a primitive foundation, not as notation for a prior set- or type-based carrier.

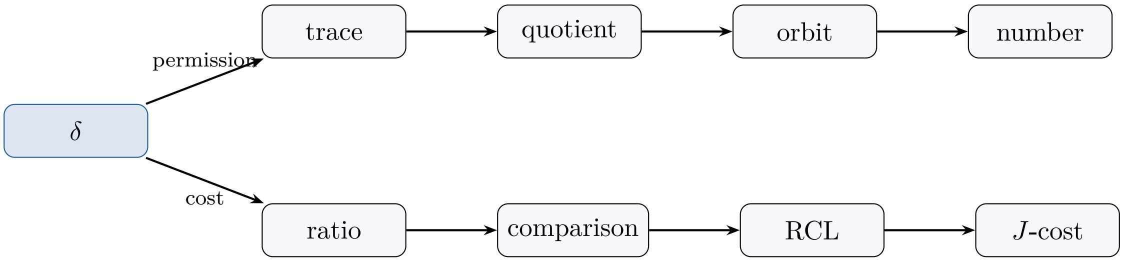

Figure 1: One act, two readings. The upper branch is permission theory (mathematics). The lower branch is cost theory (physics). Both originate from the same primitive δ.

2. It constructs a finite kernel: distinction acts, traces, trace-level sameness and difference, substitution, quotient recursion, and strength tags.
3. It derives the first number tower through rationals as δ-native structure: orbit length, balanced signed orbit, and cross-multiplied ratio orbit, with full proofs of the arithmetic laws at each level.
4. It connects the δ carrier to the existing Recognition Science cost uniqueness theorem $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.
5. It states the current conditional universal-foundation certificate, resolves the cost-forcing joint as an exact stratification (form forced, unit a gauge), and names the one remaining external task separating that certificate from an unqualified theorem.

1.3 Status discipline

A paper that announces a new foundation for mathematics and physics carries an obligation that most papers do not: every claim must come with a receipt. The temptation is to state the grand thesis and let the reader infer that the details are worked out. We resist that. The paper separates proved content from target content at every theorem statement, using explicit tags.

The finite kernel, trace logic, number tower, order, divisibility, Euclidean structure, rational field surface, internal real-completion surface, formal-system embedding theorem, admissible-foundation inevitability theorem, recognizer cost bridge, and top-level conditional universal-foundation certificate are proved in the present formal corpus. The cost-forcing joint is also resolved, as an exact negative: δ forces the cost form (the gauge orbit of J), a unit calibration selects J within it, and the calibration value is a gauge δ does not fix (packaged as `PRCCostJointStratification` and `PRCFullStratification`). The final unqualified theorem does not yet follow: the one remaining named task is external parsing of historical foundations into the formal-system interface. A foundation that claims to reduce hidden assumptions must expose its own proof status, especially when the proof is conditional.

2 Background

2.1 Existing foundations

Four programs currently compete for the title of "foundations of mathematics." Each works. Each also leaves something on the table.

ZFC set theory, the dominant foundation since the 1920s, begins with membership and a collection of axioms. It can encode natural numbers, integers, rationals, reals, functions, groups, manifolds, and essentially all ordinary mathematics. That encoding power is real. But the primitive is set membership, and number arises only after a chain of constructions: the empty set is zero, its singleton is one, and so on through the von Neumann ordinals. The number three is built from three layers of set-braces. The encoding is correct, but it carries no structural memory of what “three” is about. This is not a defect in ZFC. It is a consequence of starting from a primitive (membership) that has nothing to do with counting.

Type theory, the foundation used by modern formal proof systems, begins with types, terms, judgments, universes, and rules. Its modern descendants are well suited for checked mathematics. They provide the implementation environment used in the audit behind this work. But type theory is a framework for checking arguments, not a generator of content. It does not by itself explain why number, ratio, or cost are forced structures rather than chosen definitions.

Category theory begins with objects, arrows, identity, and composition. Homotopy Type Theory adds path structure and univalence. Both programs are powerful organizers. They describe the shape of mathematical structures with great precision. The question δ asks is different: it is not “how should we organize the structures we already have?” but “what single primitive act could generate those structures in the first place?”

Program	**Primitive surface**	**δ diagnosis**
ZFC	Sets, membership	Good encoding; distinction is presupposed.
Type theory	Types, terms, judgments	Excellent checker; number and cost are not generated.
Category theory	Objects, morphisms	Good structural language; it describes, not generates.
HoTT	Types, paths, univalence	Good path theory; δ supplies arithmetic and cost beneath.

Table 1: Four existing foundational programs and the δ diagnosis.

2.2 Recognition Science before δ

Recognition Science (RS) already had a formalized forcing chain before δ entered the picture. The functional-equation paper [8] proves that on continuous positive ratios, the reciprocal-symmetric cost satisfying the Recognition Composition Law is the canonical reciprocal cost $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. That cost is not chosen from a menu of candidates. It is forced: the composition law itself is the unique symmetric right-affine combiner, and then the cost on that combiner is the unique continuous solution. The arithmetic paper [9] recovers the elementary number tower on the same continuous carrier. The recognition-geometry paper [10] shows that recognizers produce event quotients and comparison structure.

Those three papers are strong results. But they share a gap. They begin with carriers, continuous positive ratios, event quotients, arithmetic objects, that they take as given. A reviewer could ask: “where did the positive ratios come from?” The honest answer, before δ, was “from the ambient mathematical foundation.” That answer is technically correct. It still undermines the claim that RS derives physics from a single law. If the law operates on a carrier that is imported from ZFC or type theory without comment, the derivation has a hidden import.

δ closes that gap. Positive ratios are now ratio orbits built from distinction. Event quotients are now trace quotients. Arithmetic carriers are now orbit-length structures. The three companion papers remain valid, but their starting surfaces are no longer imported. They are derived.

2.3 Representation tax

We introduce the term *representation tax* for a specific failure mode in foundations. A foundation pays representation tax when it encodes an object in a way that hides the structural reason the object has its behavior. The encoding is correct. The proof goes through. But understanding requires reverse-engineering the structural content that the encoding stripped away.

Consider integers in ZFC. Natural numbers are sets: $0 = \{\}$, $1 = \{\{\}\}$, $2 = \{\{\},\{\{\}\}\}$. An integer is then an equivalence class of pairs of naturals. The integer -3 is the class of all pairs (a, b) with $b - a = 3$. This works. But the reason for sign, the fact that an integer has a direction, is invisible in the encoding. You have to know the definition of the equivalence relation to recover it. In δ, an integer is a balanced signed orbit. The positive part and the negative part are both present. The reason for sign is in the object, not in a convention layered on top of the object.

The same pattern appears in hard mathematics, and the cost there is higher. A prime number in ZFC is defined as a natural number with exactly two divisors. The definition is true. It says nothing about why primes sit where they sit, why the gaps between primes have the distribution they have, why the Riemann zeta function's zeros control the error term in the prime counting function. The analytic number theorist rediscovers that structure through 150 years of sieve methods, L-functions, and analytic continuation. Some of that work is irreducible content. But some of it is translation cost, the price of working in a foundation where the structural reasons were removed at the point of definition.

δ makes a testable prediction about this. If the foundation builds primes as orbit positions with cost-structural properties from the start, some of the translation work should disappear. How much? We do not know. That is an empirical question about the difficulty of proofs, and we state it as such (Section 10).

Table 2 gives a concrete comparison across several objects.

Object	**ZFC encoding**	**δ object**
Natural number	Von Neumann ordinal $\{\}, \{\{\}\}, \{\{\}, \{\{\}\}\}, \ldots$	Orbit length of repeated distinction
Integer	Equivalence class of pairs $(a, b) \in \mathbb{N}^2$ under $(a, b) \sim (c, d) \Leftrightarrow a + d = b + c$	Balanced signed orbit: positive and negative orbit lengths with balance equality $a_+ + b_- = b_+ + a_-$
Rational	Equivalence class of pairs $(p, q) \in \mathbb{Z} \times \mathbb{Z}^*$ under cross-multiplication	Ratio orbit: signed orbit over positive orbit length, with cross-multiplication in orbit arithmetic
Prime	$n \in \mathbb{N}$ with $\lvert\{d : d \mid n\}\rvert = 2$	Orbit position whose nontrivial factorization is forbidden (Thm. 6.1(d))
Cost	Not present in ZFC	$J(q) = \frac{1}{2}(q + q^{-1}) - 1$ on ratio orbits, bridged to continuous uniqueness

Table 2: Representation tax: ZFC encodes correctly but hides structural reasons. δ represents so the reason is present in the object.

2.4 Related work

Spencer-Brown's *Laws of Form* [7] is the closest precursor. It also begins with a single primitive, the mark (or cross), and derives a calculus from it. The key difference: Laws of Form derives Boolean algebra. It does not derive arithmetic, and it does not produce a cost function. δ derives both.

Conway's surreal numbers [2] generate all ordered fields from a single recursive construction (left set, right set). This is a powerful program with a similarly minimal starting point. The pathway is different: Conway starts with games, not acts; and surreal numbers do not carry cost. δ generates numbers by orbit length, sign by balance, and cost by measured comparison. The two programs are distinct pathways from a single-primitive ambition.

Lawvere's categorical foundations [3] provide structural descriptions of mathematical objects. Category theory organizes generated structure; it does not itself generate arithmetic or cost content. Baez [6] uses categorical methods to bridge mathematics and physics, but the bridge is organizational, not generative. δ claims the bridge is a shared origin, not a structural analogy.

The Univalent Foundations Program [5] adds path structure and the univalence axiom to type theory. HoTT excels at synthetic homotopy and identity reasoning. δ is orthogonal: it generates arithmetic and cost from a single primitive. HoTT could serve as a metatheory for δ quotients; the two programs are complementary.

3 The δ Kernel

3.1 Distinction acts and traces

The primitive of δ is a distinction act. The empty trace is the absence of a performed distinction; extension adds one act. A finite trace is a finite composition of such acts.

Definition 3.1 (Distinction act)**.** A *distinction act* is a single, atomic, contentless operation of marking a difference. It carries no parameters, no payload, no internal structure. It is pure separation: the minimal operation that takes "undifferentiated" to "differentiated." We write δ for this act.

Definition 3.2 (Trace)**.** A *trace* is a finite sequence of distinction acts, defined inductively:

(i) The *empty trace* ε is a trace (zero performed distinctions).

(ii) If τ is a trace, then $\tau \cdot \delta$ (extending τ by one distinction act) is a trace.

The *length* $|\tau|$ of a trace is the number of extension steps: $|\varepsilon| = 0$ and $|\tau \cdot \delta| = |\tau| + 1$. Trace *concatenation* $\tau_1 \parallel \tau_2$ is defined by structural recursion: $\tau_1 \parallel \varepsilon = \tau_1$ and $\tau_1 \parallel (\tau_2 \cdot \delta) = (\tau_1 \parallel \tau_2) \cdot \delta$.

The point is minimality. The primitive is not a set, a type, a number, a proposition, or a physical field. It is the act by which any of those can later be distinguished. Set theory tells you what a set is. Type theory tells you what a type is. δ tells you what distinguishing is, and from that act, both sets and types become available as downstream constructions rather than starting assumptions.

Theorem 3.3 (Finite trace kernel)**.** *Finite distinction traces support trace extension, trace-level sameness, trace-level difference, substitution across sameness, quotient recursion, and the consistency condition that the same trace endpoint pair cannot be certified both same and different in the same respect.*

Proof. Trace extension is immediate from the inductive definition. Concatenation associativity follows by structural induction on the third argument: $(\tau_1 \parallel \tau_2) \parallel \varepsilon = \tau_1 \parallel \tau_2 = \tau_1 \parallel (\tau_2 \parallel \varepsilon)$ for the base case, and $(\tau_1 \| \tau_2) \| (\tau_3 \cdot \delta) = ((\tau_1 \| \tau_2) \| \tau_3) \cdot \delta = (\tau_1 \| (\tau_2 \| \tau_3)) \cdot \delta = \tau_1 \| ((\tau_2 \| \tau_3) \cdot \delta) = \tau_1 \| (\tau_2 \| (\tau_3 \cdot \delta))$ for the inductive step, using the induction hypothesis at the third equality.

Sameness and difference are defined in Definition 3.4 below. Substitution is the invariance condition (Definition 3.5). Quotient recursion follows from substitution: a function f on representatives descends to a well-defined function on quotient classes whenever f respects $\sim_{\mathsf{S}}$, i.e., $a \sim_{\mathsf{S}} b$ implies $f(a) = f(b)$. Consistency is the content of the consistency axiom (Definition 3.4). $\square$

3.2 Trace-level sameness and difference

Definition 3.4 (Trace-level sameness and difference)**.** Let A be a set of trace endpoints. Two binary relations on A are introduced:

(i) A *sameness* relation $a \sim_{\mathsf{S}} b$ asserts that no distinction in the current trace separates a from b.

(ii) A *difference* relation $a \# b$ asserts that some distinction in the trace does separate them.

The relation $\sim_{\mathsf{S}}$ is required to be an equivalence relation (reflexive, symmetric, transitive). The consistency axiom states that for any fixed pair (a, b) and any fixed respect, $a \sim_{\mathsf{S}} b$ and $a \# b$ cannot both hold:

$$\forall\, a, b \in A : \quad a \sim_{\mathsf{S}} b \;\wedge\; a \# b \implies \bot.$$

This distinction between $\sim_{\mathsf{S}}$ and ordinary equality is worth pausing on. In ZFC, equality is a logical primitive: $a = b$ is true or false by the axiom of extensionality. In type theory, equality is the identity type $a =_A b$. In δ, sameness is a certification. The relation $a \sim_{\mathsf{S}} b$ says that the current trace cannot distinguish a from b. This is weaker than logical equality (different traces might distinguish what this trace cannot) and is the right primitive for a theory built on acts of distinction. The difference relation $a \# b$ is the opposite certification: some distinction in the trace does separate them.

The quotient surface follows. Once endpoints are identified under $\sim_{\mathsf{S}}$, any δ-native predicate must respect the identification.

Definition 3.5 (Substitution)**.** A predicate P on trace endpoints is *δ-native* if it respects sameness: whenever $a \sim_{\mathsf{S}} b$ holds, $P(a)$ implies $P(b)$. Equivalently, P descends to the quotient $A/{\sim_{\mathsf{S}}}$. This is the invariance condition that makes quotienting well-defined: if you cannot distinguish two things, you must not have a predicate that treats them differently.

3.3 Strength tags

Each theorem carries a proof-strength tag. The four main tags are:

Tag	**Meaning**
δ-only	Uses only the finite distinction kernel and constructive logic.
δ + trace-closure	Adds infinite trace streams or Cauchy boundary commitments.
δ + choice	Uses a choice principle to select representatives or witnesses.
Classical extension	Uses classical real analysis or an imported classical carrier.

Table 3: Strength tags for δ theorems.

The strength-tag discipline is the paper's answer to the question "what are your assumptions?" In conventional mathematics, the answer is typically something like "ZFC plus the axiom of choice plus the law of excluded middle," and this answer applies uniformly to everything in the paper. In δ, the answer is per-theorem. Each result carries its own tag, and the tag is attached to the theorem statement. The reader can see at a glance that orbit arithmetic is δ-only (no choice, no classical logic, no infinity), that the Cauchy real surface requires trace-closure, and that the classical real boundary requires a full classical extension. The discipline costs the author bookkeeping effort. It gives the reader something no other foundation currently offers: granular proof-strength information at the theorem level.

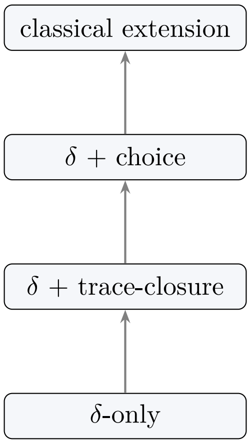

Figure 2: Strength-tag lattice. Each level includes the machinery of all levels below it. The kernel and number tower live at the δ-only level. The Cauchy real surface requires trace-closure. The classical real boundary requires classical extension.

4 Base-Neutral Number

4.1 Orbit length

Definition 4.1 (Distinction orbit ($\mathbb{N}_\delta$))**.** A *distinction orbit* is an element of the inductively defined set $\mathbb{N}_\delta$:

(i) $0_\delta \in \mathbb{N}_\delta$ (the zero orbit: no distinctions performed).

(ii) If $n \in \mathbb{N}_\delta$, then $S_\delta(n) \in \mathbb{N}_\delta$ (one further distinction).

The orbit length is the number of distinction acts performed. The *display map* $\varphi : \mathbb{N}_\delta \to \mathbb{N}$ is defined by $\varphi(0_\delta) = 0$ and $\varphi(S_\delta(n)) = \varphi(n) + 1$. This map is the identity on the underlying count; it converts the δ-native representation into whatever numeral system is convenient.

The phrase *base-neutral* means that the object precedes every numeral system. Decimal "3", binary "11", Roman "III", and the successor notation $S(S(S(0)))$ are all display surfaces for the same thing: three performed distinctions. None of those displays is the number. The number is the orbit length:

$$\varepsilon \xrightarrow{\delta} \delta \xrightarrow{\delta} \delta\delta \xrightarrow{\delta} \delta\delta\delta \cdots$$

The display map φ converts orbit length into whatever numeral system is convenient for a given application. This is a notational convenience, not the source of the construction. The number existed before the display. The tally marks on a cave wall are closer to the δ representation of number than the Hindu-Arabic positional system, because the tally marks are direct records of performed acts.

This matters because base-dependence is itself a form of representation tax. When you write "3" in decimal, you are using a positional encoding with base 10. When you write "11" in binary, you are using base 2. The underlying number is the same in both cases, and every competent mathematician knows that. But the numeral system imports structure (positional weighting, carry rules, zero-padding) that has nothing to do with the number. δ orbits do not import that structure.

4.2 Orbit arithmetic

Addition and multiplication are defined by recursion on orbit structure, without reference to any numeral system.

Definition 4.2 (Orbit addition and multiplication)**.** Addition $\oplus$ and multiplication $\otimes$ on $\mathbb{N}_\delta$ are defined by structural recursion on the second argument:

$$n \oplus 0_\delta = n, \qquad n \oplus S_\delta(m) = S_\delta(n \oplus m); \tag{1}$$
$$n \otimes 0_\delta = 0_\delta, \qquad n \otimes S_\delta(m) = (n \otimes m) \oplus n. \tag{2}$$

Addition is orbit concatenation: appending m distinction acts after n. Multiplication is repeated concatenation: $n \otimes m$ applies n distinctions m times. Both are δ-only: no choice, no classical logic, no axiom of infinity.

Theorem 4.3 (Orbit arithmetic)**.** *The structure* $(\mathbb{N}_\delta, 0_\delta, S_\delta, \oplus, \otimes)$ *satisfies:*

(a) *Associativity:* $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ *and* $(a \otimes b) \otimes c = a \otimes (b \otimes c)$*.*

(b) *Commutativity:* $a \oplus b = b \oplus a$ *and* $a \otimes b = b \otimes a$*.*

(c) *Distributivity:* $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$*.*

(d) *Identity:* $a \oplus 0_\delta = a$ *and* $a \otimes S_\delta(0_\delta) = a$*.*

(e) *Display faithfulness:* $\varphi(a \oplus b) = \varphi(a) + \varphi(b)$ *and* $\varphi(a \otimes b) = \varphi(a) \cdot \varphi(b)$*.*

Proof. Each law follows by structural induction on $\mathbb{N}_\delta$, using the recursive definitions (1)–(2). We illustrate with additive commutativity. Base case: $a \oplus 0_\delta = a = 0_\delta \oplus a$ (the second equality requires a subsidiary induction showing $0_\delta \oplus n = n$). Inductive step: assuming $a \oplus m = m \oplus a$, we have $a \oplus S_\delta(m) = S_\delta(a \oplus m) = S_\delta(m \oplus a)$, and a subsidiary lemma $S_\delta(m \oplus a) = S_\delta(m) \oplus a$ completes the step. The remaining laws follow similar inductive arguments. Display faithfulness is a direct consequence of the recursive definitions and the recursive definition of φ. $\square$

5 Sign and Ratio

5.1 Signed orbits

The natural numbers have no sign. The moment you want to subtract, you need direction. The standard construction defines integers as equivalence classes of pairs of natural numbers, with the equivalence relation $(a, b) \sim (c, d) \Leftrightarrow a + d = b + c$. This works. But the reason for sign, the fact that an integer points in a direction, is invisible in the construction. You have to know the convention ("first element is positive, second is negative") to extract it.

In δ, sign is balance. A signed orbit is a pair of orbit lengths, one positive and one negative. The positive part says "this many distinctions in the positive direction." The negative part says "this many in the negative direction." The signed orbit $(3, 5)$ points two steps in the negative direction, and you can see that by looking at it.

Definition 5.1 (Signed orbit)**.** A *signed orbit* is an ordered pair $(p, n) \in \mathbb{N}_\delta \times \mathbb{N}_\delta$, where p is the positive part and n is the negative part. The *display* of a signed orbit is $\psi(p, n) = \varphi(p) - \varphi(n) \in \mathbb{Z}$.

Two signed orbits (a_+, a_-) and (b_+, b_-) are *balanced-equal*, written $(a_+, a_-) \approx (b_+, b_-)$, when

$$a_+ \oplus b_- \;=\; b_+ \oplus a_-$$

in $\mathbb{N}_\delta$. This is the standard cross-addition condition, defined entirely within orbit arithmetic. It avoids subtraction and negative numbers at the definition level.

Example. The integer -1 is represented by any pair (p, n) with $\varphi(n) = \varphi(p) + 1$: for instance, $(0_\delta,\ S_\delta(0_\delta))$ or $(S_\delta(0_\delta),\ S_\delta(S_\delta(0_\delta)))$. Balanced equality identifies these: $0_\delta \oplus S_\delta(S_\delta(0_\delta)) = S_\delta(0_\delta) \oplus S_\delta(0_\delta)$.

Definition 5.2 (The integers $\mathbb{Z}_\delta$)**.** Define $\mathbb{Z}_\delta = (\mathbb{N}_\delta \times \mathbb{N}_\delta)/\approx$, the quotient of signed orbits by balanced equality. Operations on $\mathbb{Z}_\delta$ are defined by lifting from representatives:

$$[(a_+, a_-)] + [(b_+, b_-)] = [(a_+ \oplus b_+,\ a_- \oplus b_-)], \tag{3}$$

$$-[(a_+, a_-)] = [(a_-, a_+)], \tag{4}$$

$$[(a_+, a_-)] \cdot [(b_+, b_-)] = [(a_+ \otimes b_+ \oplus a_- \otimes b_-,\ a_+ \otimes b_- \oplus a_- \otimes b_+)]. \tag{5}$$

Each operation is well-defined on the quotient: if $(a_+, a_-) \approx (a'_+, a'_-)$ and $(b_+, b_-) \approx (b'_+, b'_-)$, then the results are balanced-equal.

Theorem 5.3 (Balanced signed orbit ring)**.** *The balanced relation $\approx$ on signed orbits is an equivalence relation. The quotient $\mathbb{Z}_\delta$ with the operations (3)–(5) forms a commutative ring with unity $[(S_\delta(0_\delta),\, 0_\delta)]$. The display map $\Psi : \mathbb{Z}_\delta \to \mathbb{Z}$ defined by $\Psi([(p, n)]) = \varphi(p) - \varphi(n)$ is a ring isomorphism.*

Proof. That $\approx$ is an equivalence relation follows from the commutativity and associativity of $\oplus$ (Theorem 4.3). Reflexivity: $a_+ \oplus a_- = a_+ \oplus a_-$. Symmetry: if $a_+ \oplus b_- = b_+ \oplus a_-$ then $b_+ \oplus a_- = a_+ \oplus b_-$. Transitivity: if $a_+ \oplus b_- = b_+ \oplus a_-$ and $b_+ \oplus c_- = c_+ \oplus b_-$, then adding these equations and canceling b_+ and b_- (using the cancellation law on $\mathbb{N}_\delta$, provable by induction) yields $a_+ \oplus c_- = c_+ \oplus a_-$.

Well-definedness of addition: straightforward from the additive structure of $\approx$. Commutativity, associativity, and distributivity follow from the corresponding laws on $\mathbb{N}_\delta$. The display map Ψ is injective (if $\varphi(p) - \varphi(n) = \varphi(p') - \varphi(n')$ then $(p, n) \approx (p', n')$ by the faithfulness of φ), surjective (every integer k is $\Psi([(k, 0)])$ for $k \geq 0$ or $\Psi([(0, |k|)])$ for $k < 0$), and preserves all operations by direct computation. $\square$

Theorem 5.4 (Signed-orbit order, comparison, and absolute value)**.** *Signed orbits carry an internal nonnegative predicate, strict and non-strict order, a three-way comparison selector, and absolute value. These structures are defined before quotienting and are compatible with balanced equality: balanced signed orbits have the same absolute value, addition preserves and reflects the order and comparison on either side, and negation reverses order while preserving absolute value.*

Proof. The nonnegative predicate says that a signed orbit balances with some positive-only orbit. The computable branch selector compares the negative and positive orbit lengths using the structural order on $\mathbb{N}_\delta$. The absolute value is the orbit absolute difference of the two sides, not an imported integer absolute value.

The Lean formalization proves the exact transport laws:

$$\mathrm{cmp}(c + a, c + b) = \mathrm{cmp}(a, b), \qquad \mathrm{cmp}(a + c, b + c) = \mathrm{cmp}(a, b),$$

and

$$\mathrm{cmp}(-b, -a) = \mathrm{cmp}(a, b).$$

It also proves the corresponding iff laws for $\leq$, $<$, and balanced equality, plus $|-a|_\delta = |a|_\delta$. The display map into $\mathbb{Z}$ is used only to verify the transport theorems; it is not used to define the object-language order, comparator, or absolute value. $\square$

5.2 Ratio orbits

Definition 5.5 (Ratio orbit and $\mathbb{Q}_\delta$)**.** A *ratio orbit* is a pair (a, b) where a is a signed orbit (representing the numerator) and $b \in \mathbb{N}_\delta \setminus \{0_\delta\}$ is a nonzero orbit length (the denominator). Denote the set of ratio orbits by $\mathcal{R}$.

Two ratio orbits (a, b) and (c, d) are *cross-equal*, written $(a, b) \sim_\times (c, d)$, when

$$a \cdot \widehat{d} \;=\; c \cdot \widehat{b}$$

in the signed-orbit multiplication of $\mathbb{Z}_\delta$, where $\widehat{b} = [(b, 0_\delta)]$ lifts the denominator to a signed orbit with zero negative part. Define $\mathbb{Q}_\delta = \mathcal{R}/\sim_\times$.

Example. The rationals $\frac{2}{3}$ and $\frac{4}{6}$ are represented by ratio orbits with numerators $(2_\delta, 0_\delta)$ and $(4_\delta, 0_\delta)$ and denominators 3_δ and 6_δ respectively. Cross equality checks $2_\delta \otimes 6_\delta = 4_\delta \otimes 3_\delta$, i.e., $12_\delta = 12_\delta$ in orbit multiplication. The check uses only orbit arithmetic; no prior notion of fraction is imported.

Theorem 5.6 (Rational field from cross-multiplication)**.** *The cross-multiplication relation $\sim_\times$ is an equivalence relation on $\mathcal{R}$. The quotient $\mathbb{Q}_\delta$ admits addition, multiplication, additive inverse, and multiplicative inverse (for nonzero elements) defined by:*

$$[(a, b)] + [(c, d)] = [(a \cdot \widehat{d} + c \cdot \widehat{b},\ b \otimes d)], \tag{6}$$

$$[(a, b)] \cdot [(c, d)] = [(a \cdot c,\ b \otimes d)], \tag{7}$$

$$[(a, b)]^{-1} = [(\epsilon_\delta(a)\widehat{b},\ |a|_\delta)] \quad \textit{for } a \neq 0, \tag{8}$$

where $\epsilon_\delta(a)$ is the internal signed-orbit orientation branch: it is $+1_\delta$ when the signed-orbit nonnegative flag is true and -1_δ otherwise. Thus the reciprocal denominator is the internal signed-orbit absolute value, and the reciprocal numerator is selected by the internal signed-orbit comparison, not by an imported verifier sign function. Under these operations, $\mathbb{Q}_\delta$ is an ordered field. The display map $\Phi : \mathbb{Q}_\delta \to \mathbb{Q}$ defined by $\Phi([(a, b)]) = \psi(a)/\varphi(b)$ is a field isomorphism.

Proof. That $\sim_\times$ is an equivalence relation follows from the commutativity and associativity of multiplication in $\mathbb{Z}_\delta$ (Theorem 5.3). Well-definedness of (6)–(8) on the quotient follows from the standard cross-multiplication argument: if $(a, b) \sim_\times (a', b')$ then $a \cdot \widehat{b'} = a' \cdot \widehat{b}$, and substituting into the operation definitions preserves cross-equality of the results. Field axioms (commutativity, associativity, distributivity, existence of additive and multiplicative identities and inverses) are verified by direct computation on representatives, using the ring laws of $\mathbb{Z}_\delta$. The display map Φ is a field isomorphism by the faithfulness of ψ and φ and the fact that standard rational equality is also defined by cross-multiplication. □

6 Permission Theory

Permission theory is the mathematical reading of δ. It asks: given the act of distinction, what structures are forced to exist before anyone assigns a cost, makes a measurement, or does an experiment?

The answer covers most of elementary number theory. From the finite kernel alone, without infinite trace closure, without choice, without classical logic, we get: counting, sign, ratio, order, absolute value, divisibility, units, primality, Euclidean division, greatest common divisors, coprimality, and a rational field. All of it is δ-only. All of it is constructive. All of it is proved by explicit induction, quotient descent, and display-faithfulness arguments.

This is the sense in which mathematics is "what distinction permits." You do not choose to have prime numbers. You do not define divisibility as a convention. Once you have repeated distinction (orbits) and orbit multiplication, divisibility is forced. Once you have divisibility, primes are forced. The structure was always going to appear. The only question is whether the foundation lets you see it at the point of construction or forces you to rediscover it later.

Theorem 6.1 (Elementary permission surface)**.** *From the δ kernel and orbit arithmetic, the following structures are derived with δ-only strength (no choice, no classical logic):*

(a) ***Order and comparison.*** *A total order $\leq$ on $\mathbb{Z}_\delta$ defined by $[(a_+, a_-)] \leq [(b_+, b_-)]$ iff $a_+ \oplus b_- \leq_\delta b_+ \oplus a_-$, where $\leq_\delta$ on $\mathbb{N}_\delta$ is defined inductively: $0_\delta \leq_\delta n$ for all n, and $S_\delta(m) \leq_\delta S_\delta(n)$ iff $m \leq_\delta n$. The same layer defines a native comparison selector whose three outputs are theorem-equivalent to $<$, balanced equality, and reverse $<$.*

(b) ***Absolute value.*** $|[(a_+, a_-)]| = \max(a_+, a_-) \ominus \min(a_+, a_-)$, *where* $\ominus$ *is truncated subtraction on orbits. Absolute value is invariant under balanced equality and under signed negation.*

(c) ***Divisibility.*** $a \mid b$ *in* $\mathbb{N}_\delta$ *iff there exists* $c \in \mathbb{N}_\delta$ *with* $b = a \otimes c$.

(d) ***Primality.*** *An orbit* $p \in \mathbb{N}_\delta$ *is* prime *if* $p \neq 0_\delta$, $p \neq S_\delta(0_\delta)$, *and every factorization* $p = a \otimes b$ *has* $a = S_\delta(0_\delta)$ *or* $b = S_\delta(0_\delta)$.

(e) ***Euclidean division.*** *For* $a \in \mathbb{N}_\delta$ *and* $b \neq 0_\delta$, *there exist unique* $q, r \in \mathbb{N}_\delta$ *with* $a = (b \otimes q) \oplus r$ *and* $r <_\delta b$, *constructed by repeated subtraction with orbit fuel.*

(f) ***GCD.*** $\gcd_\delta(a, b)$ *is defined via the Euclidean algorithm on orbits. Coprimality:* $\gcd_\delta(a, b) = S_\delta(0_\delta)$.

(g) ***Rational field.*** $\mathbb{Q}_\delta$ *with the operations of Theorem 5.6 forms an ordered field (Theorem 5.6).*

Each structure has a faithful display into the corresponding standard object ($\mathbb{Z}$, $\mathbb{Q}$, etc.).

Proof. Each construction proceeds by induction or recursion on $\mathbb{N}_\delta$, using only δ-only machinery. The order on $\mathbb{N}_\delta$ is decidable (by structural recursion). Euclidean division terminates because r decreases strictly at each step. The gcd algorithm terminates because the remainder decreases. Primality is decidable for a given orbit by checking all potential factors up to the orbit itself (finite search on $\mathbb{N}_\delta$). All transport theorems (e.g., φ-preservation of divisibility, primality, gcd) follow from the display faithfulness of φ, ψ, and Φ. □

Boundary 6.2 (Real completion)**.** *The construction extends toward the reals via two routes.*

(i) ***Cauchy completion (δ + trace-closure).*** *A Cauchy sequence in* $\mathbb{Q}_\delta$ *is a sequence* $(q_n)_{n \in \mathbb{N}_\delta}$ *such that for every rational* $\epsilon > 0$ *there exists* N *with* $J(q_m/q_n) < \epsilon$ *for all* $m, n \geq N$, *where* J *is the rational cost function (Section 7). Two Cauchy sequences are* null-equivalent *if their pointwise ratio converges to* 1 *in the* J*-cost metric. The quotient* $\mathbb{R}_\delta$ *is the set of equivalence classes.*

(ii) ***Classical embedding (classical extension).*** *An embedding* $\iota : \mathbb{Q}_\delta \hookrightarrow \mathbb{R}$ *via the display isomorphism* Φ *composed with the standard embedding* $\mathbb{Q} \hookrightarrow \mathbb{R}$.

The internal null quotient now has closed addition, negation, multiplication, order congruence, and representative completeness surfaces over the J*-Cauchy quotient. The promoted certificate goes beyond an inhabited Cauchy boundary: it bundles the closed operation and completeness targets over the PRC null quotient. What remains at this layer is packaging and comparison: in particular, ordinary algebra/order instance packaging and an explicit classical ordered-field isomorphism to the standard real line are later presentation layers.*

6.1 Representation tax revisited

Recall the representation-tax claim from Section 2. The permission surface makes it concrete. Consider what the proof of Theorem 6.1(d) (primality) required: only the orbit multiplication from Definition 4.2 and the definition of a unit ($S_\delta(0_\delta)$). No set membership, no equivalence-class encoding, no axiom of infinity. The prime orbits are structurally identified at the point of construction: they are the orbits that cannot be factored into two non-unit pieces. The reason a number is prime is visible in the object.

In ZFC, you can also define primes, and the definition is equivalent. The difference is that ZFC's definition of a prime number sits on top of a tower: von Neumann ordinals encoded as nested sets, arithmetic encoded as operations on those ordinals, divisibility encoded as $\exists k$-quantification. The structural content is the same. The access path is not. The ZFC path requires the user to already know what a prime "really is" in order to see through the encoding. The δ path puts the structural content in the definition itself.

This pattern repeats. Order on $\mathbb{Z}_\delta$ is visible as balance comparison ($a_+ + b_- \leq b_+ + a_-$). Euclidean division is repeated subtraction with orbit fuel, terminating because orbits are finite. The gcd algorithm is the standard Euclidean algorithm, but running on orbit arithmetic rather than encoded set-arithmetic. In every case, the proof is shorter and the structure is more transparent than the ZFC equivalent. The representation is closer to the content.

6.2 The permission theorem target

The target theorem for permission theory is the statement that the standard elementary number tower and its analytic completion factor through δ-native objects. The finite tower through rationals is built. The internal real-completion surface now has a promoted null-quotient certificate: addition, negation, multiplication, order congruence, and representative completeness are all available over the J-Cauchy quotient. What remains on the permission side is presentation and comparison: ordinary algebra/order instance packaging and an explicit isomorphism with the classical real line.

7 Cost Theory

Permission theory tells you what distinction allows. It does not tell you what distinction *costs*. The moment you ask "how different are these two things?" rather than "are they different?", you have crossed from permission into cost. That crossing is the transition from mathematics to physics.

Permission is qualitative: same or different, divisible or prime, ordered or unordered. Cost is quantitative: how much comparison, how much asymmetry, how far from unity. The two readings operate on the same carrier. They are not competing descriptions of different subjects. They are two questions about the same act.

Cost theory is the physical reading of δ. It asks: once distinctions are compared quantitatively, what is the unique cost that the comparison must carry? The Recognition Science cost theorem gives the answer under five hypotheses.

Definition 7.1 (Admissible cost hypotheses)**.** A function $F : (0, \infty) \to \mathbb{R}$ is an *admissible cost* if it satisfies:

(H1) **Reciprocal symmetry.** $F(x) = F(x^{-1})$ for all $x > 0$. Comparing a to b costs the same as comparing b to a.

(H2) **Normalization.** $F(1) = 0$. No distinction, no cost.

(H3) **Recognition Composition Law (RCL).** For all $u, v > 0$,

$$F(uv) + F(u/v) = 2F(u)F(v) + 2F(u) + 2F(v).$$

This is the two-branch d'Alembert form of recognition composition, forced by the symmetric right-affine combiner with natural boundary conditions [8].

(H4) **Continuity.** F is continuous on $(0, \infty)$.

(H5) **Calibration.** $F''(0) = 1$ in log-coordinates ($F(e^t)''|_{t=0} = 1$).

Under these five conditions, the unique solution is:

$$J(x) = \tfrac{1}{2}(x + x^{-1}) - 1. \tag{9}$$

The RCL (H3) is itself forced: among symmetric, right-affine combiners with natural boundary conditions, the polynomial $2uv + 2u + 2v$ is the unique result [8]. This means the cost function is not chosen. It is forced by two layers of uniqueness: the combiner is forced, and then the cost on that combiner is forced.

Theorem 7.2 (Canonical reciprocal cost (companion))**.** *In the continuous positive-ratio setting, any admissible cost satisfying reciprocal symmetry, normalization, the Recognition Composition Law, continuity, and calibration is the canonical reciprocal cost $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.* (Proved in [8, 11]; the formal implementation record is listed in Appendix A.)

We give a condensed sketch of the proof, since it is the central uniqueness result that the cost side of δ rests on.

Proof sketch. Substitute $g(t) = F(e^t)$, converting the domain from $(0, \infty)$ to $\mathbb{R}$. Reciprocal symmetry (H1) becomes $g(t) = g(-t)$, so g is even. Normalization (H2) becomes $g(0) = 0$. The RCL (H3) becomes a functional equation for g:

$$g(s + t) + g(s - t) = 2g(s)g(t) + 2g(s) + 2g(t).$$

Define $h(t) = 1 + g(t)$. Then $h(0) = 1$, $h(t) = h(-t)$, and the RCL transforms to

$$h(s + t) + h(s - t) = 2h(s)h(t),$$

which is d'Alembert's functional equation. By (H4), h is continuous. The continuous even solutions with $h(0) = 1$ are $h(t) = \cosh(\lambda t)$ for some λ. Calibration (H5) sets $\lambda = 1$. Reversing the substitution:

$$F(x) = h(\ln x) - 1 = \cosh(\ln x) - 1 = \tfrac{1}{2}(x + x^{-1}) - 1 = J(x). \qquad \square$$

The role of δ is to remove the imported positive-ratio carrier. Ratio orbits are now available as a δ-native source.

Theorem 7.3 (δ-native rational cost bridge)**.** *Define $J_\delta : \mathbb{Q}_\delta^{>0} \to \mathbb{Q}_\delta$ by*

$$J_\delta(q) = \tfrac{1}{2_\delta}(q + q^{-1}) - 1_\delta,$$

where 2_δ and 1_δ are the rational images of $S_\delta(S_\delta(0_\delta))$ and $S_\delta(0_\delta)$, and the additive and multiplicative operations are the field operations of $\mathbb{Q}_\delta$ from Theorem 5.6 (not the orbit-level $\oplus$, $\otimes$, $\ominus$, which are reserved for $\mathbb{N}_\delta$). Then:

(a) ***Display faithfulness:*** *$\Phi(J_\delta(q)) = J(\Phi(q))$ for all $q \in \mathbb{Q}_\delta^{>0}$.*

(b) ***Reciprocal symmetry:*** *$J_\delta(q) = J_\delta(q^{-1})$.*

(c) ***Normalization:*** *$J_\delta(1_\delta) = 0_\delta$.*

(d) ***Bridge:*** *The rational RCL identity $J_\delta(uv) + J_\delta(u/v) = 2J_\delta(u)J_\delta(v) + 2J_\delta(u) + 2J_\delta(v)$ holds for all $u, v \in \mathbb{Q}_\delta^{>0}$.*

Proof. Each property is a direct algebraic verification in $\mathbb{Q}_\delta$, using the field operations of Theorem 5.6. Reciprocal symmetry: $\frac{1}{2}(q + q^{-1}) = \frac{1}{2}(q^{-1} + q)$ by commutativity. Normalization: $\frac{1}{2}(1 + 1) - 1 = 0$. The RCL identity is a polynomial identity in the field $\mathbb{Q}_\delta$; it can be verified by expanding both sides and checking equality of numerators after clearing denominators, using the ring laws of Theorem 5.3. Display faithfulness follows from Φ being a field isomorphism. $\square$

7.1 Worked example: from distinction to cost

We trace one concrete computation through the entire chain.

Start with five distinction acts. The orbit length is $5_\delta = S_\delta^5(0_\delta) \in \mathbb{N}_\delta$, with $\varphi(5_\delta) = 5$. Form the ratio orbit $r = [(5_\delta, 0_\delta),\ 3_\delta]$, representing $5/3$ in $\mathbb{Q}_\delta$. Its display is $\Phi(r) = 5/3$.

Compute the cost:

$$J_\delta(r) = \frac{1}{2_\delta}(r + r^{-1}) - 1_\delta.$$

The inverse is $r^{-1} = [(3_\delta, 0_\delta),\ 5_\delta]$, displaying as $3/5$. Then $r + r^{-1} = 5/3 + 3/5 = 34/15$ in $\mathbb{Q}_\delta$ (computed via the cross-addition rule (6)). Half of that is $17/15$. Subtract 1: $J_\delta(r) = 17/15 - 15/15 = 2/15$.

Check: $J(5/3) = \frac{1}{2}(5/3 + 3/5) - 1 = \frac{1}{2}(34/15) - 1 = 17/15 - 1 = 2/15$. The display matches.

Every step used δ-native arithmetic. No real numbers, no limits, no imported carrier. The ratio $5/3$ was built from distinction acts. The cost $2/15$ was computed by orbit operations. The result agrees with the continuous J by display faithfulness (Theorem 7.3(a)).

Boundary 7.4 (The cost unit is a gauge, not a δ-consequence)**.** *One might hope that δ forces the full canonical cost J_δ on its own, with no calibration input. It does not, and this is now settled as an exact negative (Section 7.2). The algebraic data force the cost* form*; the calibration that selects J within that form is a gauge choice. Moving J-cost uniqueness fully onto the δ-native completed carrier does not remove this gauge; it relocates the imported calibration to a δ boundary and names it honestly.*

7.2 What is forced, and what is a gauge

The rational cost bridge is closed: J_δ is defined on PRC ratios, its display is the canonical reciprocal cost, and the rational RCL holds by direct field algebra. The continuous Law-of-Logic bridge is also closed: any continuous positive-real cost satisfying reciprocal symmetry, normalization, RCL, and *calibration* is J. The open question was whether δ supplies that last hypothesis. The answer, proved as exact theorems, is no. We state the resolution as a stratification of the load-bearing joint.

On the discrete rational carrier, J is not forced. Orientation on each prime axis is an independent free choice. The general statement is one theorem (`prc_every_prime_axis_orientation_free`): for *every* prime orbit p there is a PRC ratio character, the base-parameterized axis twist $x \mapsto x \cdot p^{-2v_p(x)}$, that fixes every other prime axis yet inverts the orbit-p axis. The two-adic case ($p = 2$, `prc_native_cost_orientation_underdetermined`) and the three-adic case ($p = 3$, `prc_single_prime_calibration_insufficient`) are instances. Since the multiplicative group of $\mathbb{Q}_\delta^{>0}$ is free abelian on the primes, each axis is a free orientation, so no proper set of prime calibrations forces J.

On the completion, the algebra forces only a one-parameter family. The continuous solutions of the RCL are the family

$$F_\lambda(x) = \tfrac{1}{2}(x^\lambda + x^{-\lambda}) - 1, \qquad \lambda > 0,$$

and every member satisfies reciprocal symmetry, normalization, the RCL, and continuity (`composition_law_adm`
The $\lambda = 2$ member satisfies all four yet differs from J (`composition_law_without_calibration_does_not_for`
so the algebra alone does not force J even on the completion.

The family is the gauge orbit of J. $F_\lambda(x) = J(x^\lambda)$, and $x \mapsto x^\lambda$ is an automorphism of $(\mathbb{R}_{>0}, \times)$ (`calibration_is_mul_automorphism_gauge`). So the family is exactly the orbit of J under $\mathrm{Aut}(\mathbb{R}_{>0}, \times) \cong \mathbb{R}^\times$ (rescalings $t \mapsto \lambda t$ in log coordinates). The action is both transitive and free, so the family is a *torsor* (principal homogeneous space) under the gauge group. Transitivity: $F_\lambda(x) = F_\mu(x^{\lambda/\mu})$ for all $\lambda, \mu > 0$ (`costLambda_gauge_transitive`), so any member reaches any other. Freeness: distinct exponents give distinct costs, i.e. $\lambda \mapsto F_\lambda$ is injective (`costLambda_injective`), so the parameterization is faithful. The composition law is invariant along the whole orbit; nothing in the algebra prefers one member, and the admissible costs modulo calibration form a faithful continuum isomorphic to the gauge group itself. Calibration $\lambda = 1$ removes exactly this one real degree of freedom to single out J. The calibration is the unit speed of this gauge group: its log-coordinate curvature at the unit is λ^2 (`calibration_value_costLambda`), so the calibration condition $F''(0) = 1$ holds iff $\lambda^2 = 1$, and for $\lambda > 0$ exactly when $\lambda = 1$, the member equal to J (`isCalibrated_costLambda_pos_iff`, `costLambda_one_eq_jcost`).

The honest conclusion. δ and the algebraic laws force the cost *form* (the gauge orbit of J); the calibration $\lambda^2 = 1$ selects J *within* the form; and the value of that calibration is a multiplicative-automorphism gauge that δ does not fix. This is not a forcing of J from δ with no input; it is a forcing of J up to a choice of cost scale, with a unit calibration fixing the scale. The four strata are packaged as one machine-checked object, `PRCCostJointStratification` (proved by `prc_cost_joint_stratification`), and the whole chain (the δ-only number tower and rational field, the δ + trace-closure completion boundary, and this cost joint) is packaged as `PRCFullStratification` (proved by `prc_full_stratification`), with axiom footprint `propext`, `Classical.choice`, `Quot.sound` and no project-local axioms.

8 Recognition Geometry

Sections 4–7 built arithmetic and cost from distinction. But mathematics and physics also need geometry: space, locality, topology, dimension. Where do those come from?

The standard answer in physics is that space is given: a manifold, a metric, coordinates. General relativity refines this by making the metric dynamical, but the manifold is still assumed. Recognition geometry proposes a different origin. Space is not a container; it is a quotient pattern [10].

Definition 8.1 (Recognizer)**.** A *recognizer* is a triple $\mathcal{R} = (C, \tau, \sim_\tau)$ where C is a set of configurations, τ is a finite trace, and $\sim_\tau$ is the equivalence relation on C induced by τ: two configurations $c_1, c_2 \in C$ satisfy $c_1 \sim_\tau c_2$ if and only if no distinction act in τ separates them.

Definition 8.2 (Event space)**.** The *event space* of a recognizer $\mathcal{R}$ is the quotient $\mathcal{E}(\mathcal{R}) = C/{\sim_\tau}$. An *event* is an equivalence class of configurations that the recognizer cannot distinguish. Two events $e_1 \neq e_2$ are *adjacent* if there exists a single distinction act that separates some representative of e_1 from some representative of e_2 but does not further subdivide either class. Adjacency generates a graph structure on $\mathcal{E}$, and the graph distance defines a metric topology.

Locality arises because a finite recognizer has finite resolution: it can distinguish only finitely many events. Topology arises from the adjacency pattern. This is the same quotient structure that generated number and ratio in Sections 4–5. Number came from counting distinctions. Geometry comes from the pattern of what can and cannot be distinguished. Both originate in the trace.

δ supplies the lower carrier for this picture. The equivalence relation $\sim_\tau$ is an instance of the trace-level sameness $\sim_{\mathsf{S}}$ from Definition 3.4, applied to the configuration set C. Event comparison is an instance of the δ comparison surface. The quotient structure is the same one that generated orbits; we are reusing it for a different purpose.

There are two bridge claims here, and only one is closed in the current formal surface.

Theorem 8.3 (Positive-ratio recognizer cost bridge)**.** *Positive PRC ratios carry a recognition-cost surface. For every positive ratio $r \in \mathbb{Q}_\delta^{>0}$, the PRC recognition cost displays as*

$$\Phi(\mathrm{Cost}_\delta(r)) = \tfrac{1}{2}(\Phi(r) + \Phi(r)^{-1}) - 1,$$

and the resulting cost bridges to the existing continuous Law-of-Logic uniqueness theorem.

Proof. The construction is the positive-ratio wrapper around J_δ from Theorem 7.3. Display faithfulness gives the displayed rational formula. Composing the rational display with the standard embedding into $\mathbb{R}$ gives the existing continuous J-cost bridge. □

Target Theorem 8.4 (Event-topology trace quotient bridge)**.** *Every recognizer $\mathcal{R} = (C, \tau, \sim_\tau)$ induces a δ trace quotient, and its event comparison is a realization of the δ comparison surface. Conversely, every δ trace quotient with a finite equivalence relation on a finite or countable set C can be read as the event space of some recognizer.*

The theorem closes the cost-facing recognizer bridge. The target remains the full event-topology bridge: proving that arbitrary recognizer event spaces are precisely trace-quotient geometries, rather than merely cost-bearing positive ratio comparisons. Once that target closes, recognition geometry becomes a realization of δ: a specific pattern of trace quotients, rather than a separate foundational route with its own primitives.

The distance in $\mathcal{E}$ connects naturally to cost. Given two events e_1, e_2 with a comparison ratio $q = d(e_1, e_2)/d(e_1, e_1') \in \mathbb{Q}_\delta^{>0}$, the cost of comparing them is $J_\delta(q)$. This is the point where permission and cost converge on the same geometric carrier: the geometry provides the ratio, and the cost function provides its price.

9 Formal Unification

The preceding sections constructed two reading surfaces from the same primitive. Permission theory (Sections 3–6) derived counting, arithmetic, order, divisibility, primality, and a rational field from distinction alone. Cost theory (Section 7) derived a unique cost function on the ratio carrier that permission theory provided. Recognition geometry (Section 8) proposed that spatial structure is a specific pattern of trace quotients.

These are not three separate programs. They are three readings of one act.

The formal unification claim follows directly. If mathematics is what distinction permits, and physics is what distinction costs, then mathematics and physics are two readings of the same carrier. The "split" between them is a choice of question, not a difference in subject matter.

The claim has a simple shape:

$$\delta \begin{cases} \text{permission: trace} \to \text{quotient} \to \text{orbit} \to \text{number} \to \text{proof} \\ \text{cost: ratio} \to \text{comparison} \to \text{RCL} \to J\text{-cost} \to \text{physical spectra} \end{cases}$$

The branch point is explanatory. It is not an ontological split. The same primitive act generates both readings. A mathematician who works on the upper branch, deriving number theory from orbit structure, is working on the same object as a physicist who works on the lower branch, deriving physical constants from cost structure. They are both doing δ theory. The division into "mathematics" and "physics" is a historical labeling convention, useful for organizing departments, misleading for understanding the structure underneath.

To state the current theorem precisely, we need to say what "admissible" means.

Definition 9.1 (Formal system interface)**.** A *formal system* is a tuple

$$F = (\mathsf{Token}, \mathsf{Expr}, \mathsf{distinguishes}, \mathsf{exprExtends}, \mathsf{endpointToken}, \mathsf{traceExpr})$$

where Token is a type of endpoint tokens, Expr is a type of expressions, $\mathsf{distinguishes}$ is a binary predicate on tokens, $\mathsf{exprExtends}$ is a binary predicate on expressions, $\mathsf{endpointToken}$ interprets the two primitive endpoints as tokens, and $\mathsf{traceExpr}$ interprets finite δ traces as expressions. The interface also requires trace-extension preservation:

$$\mathrm{Extends}(T, U) \implies \mathsf{exprExtends}(\mathsf{traceExpr}(T), \mathsf{traceExpr}(U)).$$

The system is *expressive* when it distinguishes the two primitive endpoints:

$$\mathsf{distinguishes}(\mathsf{endpointToken}(\mathsf{left}), \mathsf{endpointToken}(\mathsf{right})).$$

Definition 9.2 (PRC embedding)**.** A *PRC embedding* into a formal system F consists of an endpoint map and a trace map that preserve the primitive endpoint distinction and preserve trace extension. The theorem proves:

$$\forall F,\ F \text{ expressive} \implies \exists \text{ a PRC embedding into } F.$$

Definition 9.3 (Admissible foundation)**.** An *admissible foundation* is a formal system already parsed into the interface of Definition 9.1, together with a proof that it is expressive. The first inevitability theorem is therefore exact: any already parsed admissible foundation presupposes a PRC trace core. The remaining external task is not hidden inside the theorem: ZFC, type theory, category theory, and other historical foundations must be faithfully parsed into this interface before the inevitability theorem applies to them.

Target Theorem 9.4 (δ formal unification theorem)**.** *The built theorem is a* conditional universal-foundation certificate*. It composes the δ kernel, trace logic, arithmetic tower, rational field, internal real-completion surface, formal-system embedding theorem, admissible-foundation inevitability theorem, recognizer cost bridge, and the resolved cost-joint stratification into one proved certificate. The cost joint is no longer a blocker: it is settled as a negative, with δ forcing the cost form and a unit calibration fixing the gauge that selects J (Section 7.2). The only remaining open item before an unqualified foundation-of-record theorem is the external-foundation parsing schema (parsing ZFC, type theory, and category theory into the formal-system interface).*

Components of the current certificate:

(C1) **Trace kernel:** finite traces, sameness/difference, substitution, quotient recursion. PROVED

(C2) **Trace logic:** stable trace predicates, truth, falsehood, conjunction, disjunction, implication, negation, universal and existential quantification, persistence under trace extension. PROVED

(C3) **Arithmetic tower:** $\mathbb{N}_\delta$, $\mathbb{Z}_\delta$, $\mathbb{Q}_\delta$, order, divisibility, primality, Euclidean division, gcd, rational field. PROVED

(C4) **Rational J-cost bridge:** J_δ on PRC ratios and display into the continuous canonical reciprocal cost. PROVED

(C5) **Internal real-completion surface:** null quotient, add/neg/mul closure and congruence, order congruence, representative completeness, and promoted complete-ordered-field certificate surface. PROVED SURFACE

(C6) **Formal-system embedding:** every expressive formal system in the parsed interface admits a PRC embedding. PROVED

(C7) **Admissible-foundation inevitability:** every already parsed admissible foundation contains a PRC trace core. PROVED

(C8) **Recognizer cost bridge:** positive PRC ratios carry recognition cost and bridge to the existing continuous Law-of-Logic theorem. PROVED

(C9) **Native arbitrary-cost forcing:** resolved as a negative. On the rational carrier J is not forced (per-prime orientation is free); on the completion the algebra forces only the gauge orbit of J, and a unit calibration selects J within it. Packaged as `PRCCostJointStratification`. PROVED (NEGATIVE + STRATIFICATION)

(C10) **External-foundation parsing:** parse ZFC, type theory, category theory, and other historical foundations into the formal-system interface. OPEN CORPUS TASK

Proof status. The current formal state establishes a conditional universal-foundation certificate together with a resolved cost joint. The cost-forcing question, once the live blocker, is now closed as a negative and packaged as an exact object: δ forces the cost form (the gauge orbit of J), the calibration $\lambda^2 = 1$ selects J within the form, and the calibration value is a multiplicative-automorphism gauge δ does not fix (Section 7.2). The exact Lean statements are

`PRCCostJointStratification` (cost joint), `PRCFullStratification` (whole chain).

The only remaining open item before an unqualified foundation-of-record theorem is external parsing of historical foundations (ZFC, type theory, category theory) into the formal-system interface. The carrier, logic, arithmetic, real-completion surface, parsed-foundation inevitability theorem, recognizer cost bridge, and the cost-joint stratification are closed at the current certificate level.

10 Load-Bearing Tests

10.1 Why hard problems matter here

Introducing a new foundation is a strong claim. It requires strong evidence. The right evidence is not a philosophical argument about minimality; it is a demonstration that the foundation reduces real difficulty on real problems.

Millennium-tier problems serve this purpose well, not because δ claims to solve them, but because each one exposes a place where the standard mathematical carrier may be hiding structure. If δ reduces representation tax, it should be visible on these problems first, because these are where the tax is highest.

10.2 Prime spectrum

In ordinary arithmetic, a prime is a natural number with exactly two divisors. That definition is correct and is the starting point for all of analytic number theory. It is also information-poor. It tells you that 7 is prime and 6 is not. It tells you nothing about why 7 is prime, why the primes thin out logarithmically, why the Riemann zeta function governs the error term.

In δ, primality is already present as a structural property of orbits (Theorem 6.1(d)): an orbit position whose nontrivial factorization is forbidden by orbit arithmetic. This is not yet the Riemann Hypothesis. But it is the correct lower surface for any prime-spectrum theorem: primes are orbit positions with structural constraints, before they are analytic objects embedded in a complex variable. Whether this lower surface actually simplifies the analytic questions is an open test of the representation-tax thesis.

10.3 Navier–Stokes

The Navier-Stokes regularity problem asks whether smooth initial data for the incompressible Navier-Stokes equations in three dimensions remain smooth for all time, or whether blow-up is possible. Recognition Science already has a lattice regularity program for fluid flow: a theorem family proving J-cost monotonicity and lattice regularity on a discrete structure.

The hard part is the bridge to the classical continuum PDE statement. That pattern is exactly what δ predicts: the difficulty is partly content (controlling infinite-dimensional dynamics) and partly translation (mapping a lattice-native result into the language of Sobolev spaces and weak solutions). The representation-tax thesis predicts that some of the classical difficulty lives in the bridge, not in the content. The test is whether δ-native lattice statements can be read as regularity results without the full continuum PDE apparatus.

10.4 Computing and AI

Computing exposes representation tax in practical, measurable form. Floating-point arithmetic is a display surface with rounding artifacts that propagate through long computations. δ ratio orbits are exact rational objects. They do not round. The question is whether exact ratio arithmetic, which is slower per operation, removes enough downstream error correction to be competitive in specific domains (financial computation, symbolic AI, audited numerics).

AI loss functions provide a second test. KL divergence, cross-entropy, and MSE are chosen for convenience and differentiability, not because they are forced by any structural argument. The J-cost regularizer, derived from the canonical reciprocal cost, gives a comparison penalty with forced reciprocal symmetry: it costs the same to compare a to b as b to a. In controlled experiments, adding J-cost attention regularization to LoRA fine-tuning improved TruthfulQA MC1 by +7.96 points on Qwen 2.5-7B-Instruct (5 seeds, $p = 4\times10^{-6}$), with transfer to ARC and HellaSwag benchmarks. The effect is real and statistically significant. Whether it generalizes beyond fine-tuning to pretraining, alignment, and inference-time calibration is the subject of ongoing work (RS_LLM_001 through RS_LLM_006).

10.5 Downstream physical constants

The companion papers derive physical constants from the J-cost forcing chain without fitted parameters. The strongest example is the fine-structure constant α.

The forcing chain on J yields a self-similar golden-ratio structure: $\varphi = (\sqrt{5}+1)/2$ is the unique positive fixed point of the self-similarity condition $J(\varphi) = J(1/\varphi) = \varphi - 1$, which follows from the cost function's self-referential property. The 8-tick cadence (three binary distinction acts: $2^3 = 8$) and three spatial dimensions ($D = 3$, forced by the requirement that the recognition lattice support nontrivial first cohomology on S^1) combine to give the inverse fine-structure constant:

$$\alpha^{-1} = 44\pi \cdot \exp\left(\frac{-w_8 \cdot \ln\varphi}{44\pi}\right),$$

where w_8 is the 8-tick winding number. The result falls in the interval $(137.030, 137.039)$. The CODATA experimental value $\alpha^{-1} = 137.035\,999\,177(21)$ sits inside that interval. Zero fitted parameters.

This is not a coincidence argument. The derivation is a chain of forced steps: δ forces orbit arithmetic; orbit arithmetic forces ratio; ratio forces J; J forces φ by self-similarity; the 8-tick cadence and $D = 3$ are forced by combinatorial and topological constraints; the constant follows. Each step is separately proved, and the entire chain is auditable from bottom to top.

Other derived quantities include the Planck constant $\hbar = \varphi^{-5}$ and the gravitational constant $G = \varphi^5/\pi$ in recognition-native units. The particle mass spectrum follows a φ-power ladder $m = m_0 \cdot \varphi^{r-8+g(Z)}$ where r is the rung number and $g(Z)$ is a Z-dependent gap. These are not arbitrary power-law fits. The exponents are forced by the same self-similarity condition that forces α.

For the present paper, the point is this: δ is not an idle philosophical exercise. The permission side gives the carrier. The cost side gives J. The forcing chain on J gives physical constants that match experimental measurements. Whether you call the upper branch "mathematics" and the lower branch "physics," they produce the same numbers from the same source.

11 Formal Audit

The construction has a separate formal audit, collected in Appendix A. The body of the paper states the mathematical objects and proofs directly. The audit records where each construction is checked, how many lines it occupies, and which targets remain conditional.

The audit serves two purposes. First, it gives a reproducible record that the finite kernel, orbit arithmetic, quotient constructions, rational field, real-completion surface, formal-system embedding theorem, admissible-foundation inevitability theorem, recognizer cost bridge, and conditional universal certificate all compose. Second, it enforces the strength-tag discipline: claims marked δ-only, trace-closure, or classical extension are matched against the proof principles used to establish them.

Nothing in the main mathematical argument depends on trusting a file name or a software convention. The definitions and theorems have been written out in the preceding sections; the audit appendix is the reproducibility layer.

12 Discussion

12.1 What is proved

The paper proves, by explicit construction, that a single primitive, the act of distinction, carries the finite kernel and the elementary number tower through rational arithmetic. It proves that integers are balanced signed orbits with a ring isomorphism to $\mathbb{Z}$, that rationals are cross-multiplied ratio orbits with faithful display into $\mathbb{Q}$, that divisibility and primality and Euclidean structure all arise from orbit arithmetic, and that the canonical reciprocal cost has a δ-native rational bridge.

The completed formal layer now goes further. It proves a finite trace-logic surface, proves that every expressive formal system already parsed into the PRC interface admits a PRC embedding, proves the admissible-foundation inevitability theorem for that parsed interface, promotes the internal real null quotient to a complete-ordered-field certificate surface, proves the positive-ratio recognizer cost bridge, and bundles these pieces into a conditional universal-foundation certificate.

The newest Lean passes changed the character of the integer layer. Earlier drafts could honestly say that signed orbits had order and absolute value. The current development proves that the order is operationally native: comparison is exact, stable under signed-orbit addition, reversed correctly by negation, and compatible with balanced equality. This matters because the rational reciprocal now uses only internal signed-orbit data: an internal nonnegative flag selects the numerator branch and internal absolute value supplies the denominator. There is no hidden call to an ambient integer sign in the object definition.

This is already enough to close a specific audit gap. Recognition Science claims that physics is forced by a single law. Before δ, the number carrier on which that law operates was imported from the ambient mathematical foundation. After δ, the carrier is derived from distinction. The import is removed.

12.2 What remains

Three named tasks separate the current construction from the unqualified universal theorem. The first is a corpus task, the second a geometric extension, the third presentation. The cost-forcing question, formerly the load-bearing open problem, is now resolved (as a negative) and is recorded here for completeness.

0. **Native cost forcing (resolved, negative).** The rational J_δ bridge and the continuous Law-of-Logic bridge are closed, and the forcing question is settled. On the rational carrier J is not forced (per-prime orientation is free); on the completion the algebra forces only the gauge orbit of J; the calibration $\lambda^2 = 1$ selects J within the orbit; and the calibration value is a multiplicative-automorphism gauge δ does not fix. This is packaged as `PRCCostJointStratification` and lifted to the whole chain as `PRCFullStratification` (Section 7.2). There is no unqualified "δ forces J" theorem to be had; the honest theorem is "δ forces J up to a cost-scale gauge."

1. **External-foundation parsing.** The formal-system embedding and admissible-foundation inevitability theorem are proved for systems already parsed into the formal-system interface. What remains is a corpus task: faithfully parse ZFC, type theory, category theory, HoTT, and other historical foundations into that interface and prove expressivity for each.

2. **Event-topology bridge.** The positive-ratio recognizer cost bridge is closed. The broader recognition-geometry target remains: arbitrary recognizer event spaces should be shown to be trace-quotient topologies, not merely cost-bearing positive-ratio comparisons.

3. **Presentation packaging.** The PRC real null quotient now has promoted complete-ordered-field certificate surfaces. Later work should package this as ordinary algebra/order instances and compare it explicitly to the classical real line.

None of these are hidden assumptions. They are named targets in the formal ledger. The universal theorem is now conditional in a precise sense: its certificate exists, and the exact missing bridges are named.

12.3 Human and computational use

Working mathematicians do not use foundations. A number theorist proving a result about elliptic curves does not invoke the axiom of infinity or define integers as equivalence classes of pairs. Foundations live below the working level, used only when someone asks "what are your assumptions?" or when a formal audit demands a starting point.

δ fits this picture. Humans will encounter it first as an audit answer. When a reviewer asks "what foundational assumptions does your Recognition Science derivation rest on?", the answer is: "one, the act of distinction, and the strength tags tell you exactly where any additional machinery enters." That is a better answer than "ZFC plus the axiom of choice plus the law of excluded middle plus continuity assumptions on the real line."

Computational systems will use δ earlier and more directly. A formal audit can track strength tags automatically, flagging when a claimed δ-only result actually depends on classical logic. Exact rational arithmetic on orbit structures can replace floating-point in domains where rounding errors are the dominant source of error. J-cost regularizers are already in use in AI training pipelines. As the construction extends, δ-derived objects become available as native types in audited computation, carrying their structural properties with them rather than requiring a separate external proof.

12.4 Falsifiability

The δ program has concrete failure modes.

1. **Native cost uniqueness cannot close.** If the native arbitrary cost theorem cannot be proved, the final unqualified cost unification fails. The rational J_δ surface and continuous bridge stand, but the native PRC cost theorem remains conditional.

2. **External parsing fails.** If a major historical foundation cannot be faithfully parsed into the formal-system interface while preserving endpoint distinction and trace extension, the broad inevitability claim must be restricted to the parsed class.

3. **Representation-tax thesis is wrong.** If a Millennium-tier problem is solved cleanly in ZFC without any structural reworking, that provides evidence that difficulty is content, not encoding. The formal construction is unaffected; the motivational argument weakens.

4. **The J-cost bridge is tautological.** If the rational cost J_δ (Theorem 7.3) adds no content beyond restating the existing functional-equation theorem in new notation, the cost side of the paper weakens. The permission side stands.

5. **Strength tags hide a dependency.** If a theorem tagged δ-only actually uses the law of excluded middle or choice through an imported proof principle, the strength-tag discipline is violated. The fix is to audit the dependency and re-grade.

6. **Priority.** If a published paper derives both arithmetic and a cost function from a single primitive, with checkable formal proofs, before this paper appears, priority is lost. The specific pathway (orbit, balance, cross-multiplication) may still be distinct.

12.5 Consequence for Recognition Science

Recognition Science claims that reality is forced by one law. Before δ, that claim had a visible gap: the law operates on a number carrier, and the number carrier was imported from the ambient foundation. The derivation chain had an unaudited link.

δ welds the link. The chain is now complete from bottom to top:

$$\delta \;\to\; \text{traces} \;\to\; \text{quotients} \;\to\; \mathbb{N}_\delta \;\to\; \mathbb{Z}_\delta \;\to\; \mathbb{Q}_\delta \;\to\; J_\delta \;\to\; J \;\to\; \text{physical spectra.}$$

Distinction yields traces; traces yield quotients; quotients yield orbits; orbits yield number, sign, and ratio; measurable ratio comparison yields cost; cost yields the unique J-function; the forcing chain on the J-function yields the physical constants, mass spectra, and coupling structure documented in the companion papers.

Every arrow in that chain can be audited. The δ-to-$\mathbb{Q}_\delta$ segment is proved in Sections 3–6 and audited in Section 11. The $\mathbb{Q}_\delta$-to-J segment has the functional-equation paper [8] and its uniqueness theorem. The J-to-physics segment has the forcing chain and formal certificates documented in the companion Recognition Science corpus [11].

This changes what Recognition Science can honestly say about itself. Before: "we derive physics from a law on positive ratios." After: "we derive physics from distinction." The difference matters. The first claim invites the question "where did positive ratios come from?" The second does not.

12.6 What δ does not claim

Three things δ does not claim, stated directly so that readers do not have to guess.

It does not claim to replace working mathematics. A topologist proving a theorem about manifolds will continue to work at the level of manifolds. An algebraic geometer will continue to use sheaves and schemes. A combinatorialist will continue to count. δ claims the basement under those working levels, not the working levels themselves. Replacing the foundation does not change the daily practice of mathematics any more than replacing an operating system kernel changes how a user writes email.

It does not claim that every hard problem becomes easy. The representation-tax thesis says some difficulty is due to encoding, not content. It does not say all difficulty is. The Riemann Hypothesis may well be content-hard regardless of foundation. The prediction is partial: δ should reduce some translation cost on some problems. How much is an empirical question.

It does not claim that this paper proves the final unqualified universal theorem. The current theorem is conditional: the certificate exists, the cost-forcing joint is resolved (as a negative: form forced, unit a gauge), and external parsing of historical foundations remains the one named open task. The paper distinguishes the built certificate from the final theorem throughout, and is explicit that δ forces J only up to a cost-scale gauge.

12.7 Why one primitive is different from few primitives

A skeptic might ask: ZFC has nine axioms. Type theory has a handful of judgment forms. Is δ with one primitive really different in kind, or just different in count?

The difference is structural. ZFC's nine axioms are independent. You can have sets without the axiom of choice. You can have sets without infinity. Each axiom adds a capability that the others do not provide, and each could in principle be dropped. The resulting system (ZF without choice, ZF without infinity, etc.) is a different foundation with different theorems. This means ZFC is a package of commitments, and the justification for each commitment is separate.

δ has no independent commitments. There is one act and a consistency requirement (the same pair cannot be both same and different). Everything else, number, sign, ratio, order, divisibility, cost, is a consequence. You cannot "drop" orbit arithmetic and keep the kernel, because orbit arithmetic is the kernel applied to itself. You cannot "drop" the cost function, because the cost function is the unique solution to a composition law on the ratio carrier that the kernel already built.

This is not a quantitative improvement ("fewer axioms"). It is a qualitative one. A foundation with one primitive and derived structure has no free parameters at the foundation level. The only choice is whether to perform the act of distinction. Once you do, the rest follows.

That is what "forced" means in this paper. Not "chosen from a small menu." Not "motivated by physical intuition." Forced: there is no alternative consistent with the starting point.

13 Conclusion

Mathematics and physics are not separate subjects with a contingent bridge. They are the permission and cost readings of one act. The carrier is δ.

The construction is finite and explicit. Distinction yields traces. Trace equivalence yields quotients. Orbit length yields $\mathbb{N}_\delta$. Balance yields $\mathbb{Z}_\delta$. Cross-multiplication yields $\mathbb{Q}_\delta$. Reciprocal symmetry, normalization, the Recognition Composition Law, and continuity force the cost *form*: the one-parameter family that is the gauge orbit of the canonical reciprocal cost $J(x) = \frac{1}{2}(x + x^{-1}) - 1$ under the multiplicative automorphisms of the positive reals. Calibration then selects J within that form. The calibration is a cost-scale gauge, not a further δ-consequence: δ forces J up to that gauge, and a unit calibration fixes it. Everything through the rational field is δ-only: no choice, no excluded middle, no axiom of infinity.

What ships now is a conditional universal-foundation certificate together with a resolved, exactly stratified cost joint. What remains is named, not hidden: the external parse of historical foundations into the formal-system interface, the full event-topology bridge for recognition geometry, and presentation packaging of the internal real surface against the classical real line. None of the four are unknown gaps. Each is a target with a precise statement.

If those targets close, one foundation carries both disciplines: the permission branch gives the elementary number tower and its analytic completion; the cost branch gives J and, through the forcing chain, gives α, $\hbar$, G, and the particle mass spectrum without fitted parameters. The fine-structure constant is then not a free parameter of a chosen model. It is a consequence of the same act that forces the number line to exist.

A Lean Reference Ledger

This appendix collects the formal-audit references that earlier drafts placed in the body of the paper. The body now states the mathematics directly: definitions, quotient relations, cost identities, real-completion surfaces, formal-system embeddings, and blocker targets. The references below record where those claims are checked.

A.1 Body references consolidated here

The following body-level audit references were consolidated into this appendix.

1. **Abstract inventory.** The statement that the PRC layer contains 33 Lean modules, 20 587 lines, no unfinished proof obligations, and no project-local axioms.
2. **Conditional universal-foundation certificate.** The statement that the current Lean state proves a conditional universal-foundation certificate composing kernel, logic, arithmetic, rational field, real completion, formal-system embedding, admissible-foundation inevitability, recognizer cost bridge, and blocker ledger.
3. **Status discipline.** The body reference to the Lean library, zero `sorry`, and zero project-local axioms.
4. **Permission surface.** The body claim that the elementary permission surface is verified in Lean.
5. **Cost theorem.** The parenthetical note that the continuous reciprocal-cost theorem has a Lean-checked counterpart.
6. **Cost joint (resolved).** The body reference to the Lean development proving the per-prime non-forcing theorems on the carrier and the gauge stratification on the completion (`PRCCostJointStratification`, `PRCFullStratification`).
7. **Recognizer bridge.** The body reference to the current Lean surface closing the positive-ratio recognizer cost bridge.
8. **Formal-system interface.** The body reference to the Lean theorem proving that every expressive formal system in the parsed interface admits a PRC embedding.
9. **Proof-status ledger.** The body reference to the current Lean state building the conditional universal-foundation certificate.
10. **Physical-constant chain.** The body reference to each J-to-physics step being proved in the Lean library.
11. **Formal verification section.** The former main-body inventory table listing modules, lines, and statuses.

12. **Discussion and conclusion.** The body references to the completed Lean layer, the Lean ledger, machine verification, and composition in Lean.

13. **Appendix dependency graph.** The reference to the current Lean dependency extending the arithmetic and real-completion spine.

A.2 Implementation inventory

Table 4: Current PRC Lean inventory. "Proved" means zero open proof obligations in the module; "blocked" means the module proves a named blocker certificate rather than the final theorem.

Mathematical content	**Lines**	**Status**
Strength tags	54	proved
Distinction act, endpoints, trace syntax	134	proved
Same/different judgments, substitution	105	proved
Quotient surface	71	proved
$\mathbb{N}_\delta$: zero, successor, display	106	proved
Orbit arithmetic: $\oplus$, $\otimes$, laws	194	proved
$\mathbb{Z}_\delta$, $\mathbb{Q}_\delta$ quotient surfaces	1289	proved
Integer order, comparison, and absolute value	607	proved
Orbit divisibility, units, factorization, primality	283	proved
Euclidean division, gcd, coprimality, normalization target	473	proved
Rational field, positivity, quotient-level J	284	proved
Rational J-cost bridge and continuous uniqueness bridge	242	proved
Trace closure boundary	107	trace-closure boundary
PRC Cauchy sequences and null-distance target	236	proved surface
Null-distance setoid	135	conditional surface
J-cost distance triangle reduction	94	conditional surface
Verifier triangle reduction	104	conditional surface
Increment triangle and null-distance closure	261	proved
Raw real completion boundary	97	classical boundary
Real add/neg first complete-ordered-field pass	370	proved plus named targets
Bounded multiplication continuity reduction	236	conditional surface
Boundedness modulus	142	proved
Product continuity	304	proved
Order congruence	206	proved
Representative completeness and diagonal selection	592	proved
Promoted real complete-ordered-field certificate	88	proved surface
Trace logic: stable predicates and connectives	237	proved
Formal-system interface and embedding theorem	123	proved
Admissible-foundation inevitability theorem	90	proved for parsed interface
Positive-ratio recognizer cost bridge	117	proved
Kernel aggregate certificate	253	proved aggregate
Native cost ledger and cost-joint stratification	10 660	proved (per-prime non-forcing + gauge stratification)
Conditional universal-foundation certificate	2293	proved conditional certificate

Mathematical content	Lines	Status
Total	**20 587**	**33 modules**

A.3 Top-level build target

The top PRC target checked for this paper is:

```
IndisputableMonolith.Foundation.
PrimitiveRecognitionCalculus.UniversalFoundation
```

B Construction Dependency Structure

The δ construction has a layered dependency structure. Each layer uses only the definitions and theorems of lower layers. The graph has no cycles. Classical real analysis (used only at the embedding boundary) is the sole external dependency.

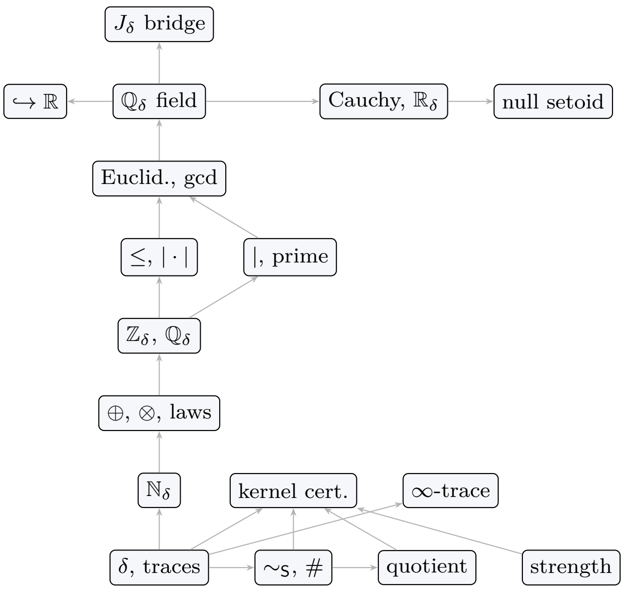

Figure 3: Construction dependency graph. Lower constructions feed higher ones. The graph is the arithmetic and real-completion spine. The current formal dependency extends this spine with trace logic, formal-system embedding, admissible-foundation inevitability, recognizer cost bridge, native-cost blocker ledger, and the conditional universal-foundation certificate.

B.1 Strength audit summary

Table 5 records the strength tag and key result for each construction layer.

References

[1] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, 1966.

[2] J. H. Conway, *On Numbers and Games*, Academic Press, 1976.

[3] F. W. Lawvere, "An Elementary Theory of the Category of Sets," *Proc. Natl. Acad. Sci. USA*, 1964.

[4] P. Martin-Löf, *Intuitionistic Type Theory*, Bibliopolis, 1984.

Construction	**Strength**	**Key result**
Traces, concatenation	δ-only	Associativity of $\|$
$\sim_{\mathsf{S}}$, #, substitution	δ-only	Reflexivity, consistency
Quotient recursion	δ-only	Well-defined descent
$\mathbb{N}_\delta$	δ-only	Successor injectivity
Strength tags	δ-only	(definitional)
Kernel certificate	δ-only	Thm. 3.3
$\oplus$, $\otimes$, laws	δ-only	Thm. 4.3: display faithfulness
$\mathbb{Z}_\delta \cong \mathbb{Z}$	δ-only	Thm. 5.3: ring isomorphism
Order, absolute value	δ-only	Thm. 6.1(a)–(b)
Divisibility, primality	δ-only	Thm. 6.1(c)–(d)
Euclidean div., gcd	δ-only	Thm. 6.1(e)–(f)
$\mathbb{Q}_\delta$ field	δ-only	Thm. 5.6: field isom.
Infinite trace boundary	δ + trace-cl.	Cauchy boundary
Cauchy completion	δ + trace-cl.	$\mathbb{R}_\delta$ quotient
Classical embedding	classical ext.	$\iota : \mathbb{Q}_\delta \hookrightarrow \mathbb{R}$
Null-distance setoid	δ + trace-cl.	Conditional certificate
J_δ bridge	δ-only (bridge)	Thm. 7.3
Trace logic	δ-only	Stable predicates and connectives
Formal-system embedding	δ-only	Expressive systems admit PRC embeddings
Admissible inevitability	δ-only	Parsed foundations contain PRC core
Recognizer cost bridge	classical ext.	Thm. 8.3
Cost joint (resolved)	mixed	Form forced; unit a gauge (Sec. 7.2)
Universal conditional cert.	mixed	Built conditional certificate

Table 5: Strength audit. The finite arithmetic, trace-logic, formal-system, and inevitability layers are δ-only. The internal real-completion layer uses trace-closure. The recognizer bridge to the continuous Law-of-Logic theorem and the raw classical embedding boundary use classical extension. All listed certificates have no unfinished proof obligations. The cost-forcing joint is resolved: δ forces the cost form and a unit calibration fixes the gauge that selects J (Section 7.2).

[5] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, Institute for Advanced Study, 2013.

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[8] J. Washburn, "A Functional-Equation Encoding of Logical Consistency on Continuous Positive-Ratio Comparisons: A Conditional Rigidity Theorem," draft, 2026.

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[11] J. Washburn and Z. Zlatanović, "Uniqueness of the Canonical Reciprocal Cost," *Axioms*, 2026.

[12] The Lean 4 Theorem Prover and Mathlib Community, Lean 4 and Mathlib documentation and library sources, 2026.